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Statistical primer: using prognostic models to predict the future: what cardiothoracic surgery can learn from *Strictly Come Dancing*

Jamie A. Mawhinney [a,*], Craig A. Mounsey [b], Alastair O'Brien [c], J. Rafael Sádaba [d] and Nick Freemantle [c]

[a] Pulvertaft Hand Centre, Royal Derby Hospital, Derby, UK
[b] Royal Berkshire Hospital, NHS Foundation Trust, Reading, UK
[c] Institute of Clinical Trials and Methodology, University College, London, UK
[d] Hospital Universitario de Navarra, Universidad Pública de Navarra, Pamplona, Spain

* Corresponding author. Pulvertaft Hand Centre, Royal Derby Hospital Kings Treatment Centre (KTC) Level 2, Uttoxeter Road, Derby DE22 3NE, UK. Tel: +44-7968-343156; e-mail: jamie.mawhinney@nhs.net (J.A. Mawhinney).

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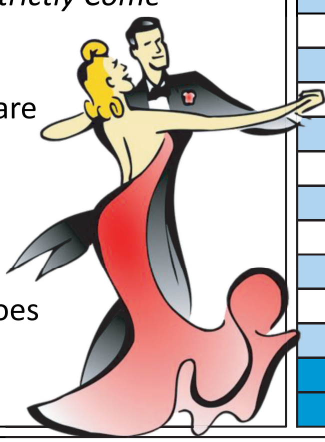
Statistical Primer: The Challenges of Prognostic Modelling

Summary

Cardiothoracic surgery commonly employs prognostic models, but challenges exist in their appropriate use. We illustrate this by forecasting the results of *Strictly Come Dancing.*

We show that predictions are improved when data are included on contestants' dance ability and demonstrate that demographic data alone does not necessarily predict the future.



Contestant	2022 Series Result	Baseline Prediction	Updated Prediction
Hamza Yassin	1	8	5
Fleur East	=2	6	7
Helen Skelton	=2	7	8
Molly Rainford	=2	1	1
Will Mellor	5	11	6
Kym Marsh	6	9	12
Ellie Taylor	7	10	10
Tyler West	8	3	4
Tony Adams	9	14	15
Ellie Simmonds	10	12	9
James Bye	11	5	11
Jayde Adams	12	4	3
Matt Goss	13	13	13
Richie Anderson	14	2	2
Kaye Adams	15	15	14
R2		0.25	0.40
Spearman's Rank Correlation		0.22	0.30

Legend: R2 = coefficient of determination.

Abstract

OBJECTIVES: Prognostic models are widely used across medicine and within cardiothoracic surgery, where predictive tools such as EuroSCORE are commonplace. Such models are a useful component of clinical assessment but may be misapplied. In this article, we demonstrate some of the major issues with risk scores by using the popular BBC television programme *Strictly Come Dancing* (known as *Dancing with the Stars* in many other countries) as an example.

METHODS: We generated a multivariable prognostic model using data from the then-completed 19 series of *Strictly Come Dancing* to predict prospectively the results of the 20th series.

RESULTS: The initial model based solely on demographic data was limited in its predictive value (0.25, 0.22; R^2 and Spearman's rank correlation, respectively) but was substantially improved following the introduction of early judges' scores deemed representative of whether contestants could actually dance (0.40, 0.30). We then utilize our model to discuss the difficulties and pitfalls in using and interpreting prognostic models in cardiothoracic surgery and beyond, particularly where these do not adequately capture potentially important prognostic information.

CONCLUSION: Researchers and clinicians alike should use prognostic models cautiously and not extrapolate conclusions from demographic data alone.

Keywords: Prognostic models · Risk prediction · EuroSCORE · STS · *Strictly Come Dancing* · Cardiothoracic surgery

ABBREVIATIONS

AIC	Akaike's information criterion
CI	Confidence interval
SCD	*Strictly Come Dancing*

INTRODUCTION

Prognostic models utilize statistical analyses of subject characteristics to describe risk and inform decision-making [1]. Such models are used in numerous fields, for example in risk profiling sex offenders and quantifying stroke and myocardial infarction risk [2, 3]. Cardiothoracic surgery is no exception. Models like EuroSCORE and the Society of Cardiothoracic Surgeons' risk calculator (STS) are widely used to estimate the risk of death or severe morbidity following cardiac surgery [4–6]. These prognostic tools were generated from large patient datasets and have been refined since their original description [7–9]. Similar strategies are used in the assessment of heart-lung transplant services in the UK [10]. When used appropriately these tools can be of great benefit to patient care, helping to support informed decision-making around potentially high-risk surgical procedures. However, such models can be misapplied, particularly when considering the difficulties of generalizing demographic data. Risk scores are only as useful as the parameters from which they are generated, and when used incorrectly can hinder clinical management and confuse potentially life-changing decisions.

To demonstrate some of the difficulties involved in developing and applying prognostic models, we have chosen to use the popular BBC One show *Strictly Come Dancing* (*SCD*) as an illustrative example. The programme first aired in 2004 and consistently receives some of the highest viewing figures of any UK television show [11]. The show has since spread internationally, with versions currently broadcast in Italy, France and Germany. Celebrities are paired with professional dancers and compete in a knockout competition where they are eliminated based on a combination of viewers' votes and the scores of an expert panel.

Following the authors' previous informal, and largely unsuccessful, attempts to forecast the outcome of *SCD*, in this report, we outline our development of a multivariable prognostic model to predict the results of the 2022 series. Although clearly very different from the world of cardiothoracic surgery, we are able to use these results to highlight some of the key difficulties in the appropriate use of risk prediction scores and draw inference for the utility of prognostic models in this field and beyond.

MATERIALS AND METHODS

A glossary summarizing the specific statistical terminology we have used can be found in Table 1.

Ethics statement

All data were publicly obtainable and therefore ethics approval was not necessary.

'Strictly Come Dancing' format

Despite its popularity, not all readers may be aware of the intricacies of *SCD* and hence some description of the format of the show is necessary. There has been some variability throughout the history of the show, but the basic format remains the same. Each celebrity is paired with a professional dance partner for the duration of the series. They then compete in a knockout competition and perform a different dance each week which is scored by a four-judge panel. After all the contestants have danced, they are ranked based on a combination of the summed judges' scores and viewer's votes. The lowest ranking 2 couples then compete in a 'dance off', after which the judges choose which couple will leave the competition. In this way, couples are sequentially eliminated from the competition until only the series champion (and their professional dance partner) remain.

Data collection and variables

A comprehensive dataset, analogous to any centrally collated data registry, was compiled using freely available data and included all contestants from the main *SCD* series (1–20). Data were entered by 1 researcher and checked by a second. Outcome data were available for the first 19 series at the point of analysis and therefore the model was generated from this data to predict the results of series 20. Initially, we set out to attempt to predict the results of the show based purely upon demographic data. These were selected *a priori* and included age, sex, profession, ethnicity, professional partner and series as the principal explanatory variables. Age was included as a continuous variable, and ethnicity was simplified

Table 1: Glossary of statistical terminology in order of use within the text

Terminology	Definition
Poisson mixed model	Poisson models are used where the response variable is a count. They are often used with mixed (random) effects which model structures in the data (such as belonging to the same geographical area) efficiently.
Binomial mixed model	Binomial models are used where the response variable is in the form of event/non-event (such as dead/alive). They are often used with mixed (random) effects which model structures in the data (such as belonging to the same geographical area) efficiently.
Continuous model	Continuous models, also sometimes known as analysis of variance, are used when the response variable can be summarized as a difference in mean values.
Cox constant proportional hazards survival model	Cox constant proportional hazards survival models are a type of survival model. Survival models describe the relationship between time passing and 1 or more covariates before an event occurs. In a proportional hazards model, the relationship between the explanatory variables and the end point is constant on a ratio scale.
Restricted cubic spline with 5 knots	Restricted cubic splines enable a statistical model to model explanatory variables with a non-linear (curved) relationship with the response variable.
AIC	AIC is a measure of relative model fit. The value from a model has no direct interpretation, but it is useful for comparing nested models (that is for otherwise identical models where one has a subset of the explanatory variables included in the other). A lower value demonstrates improved model fit. A reduction of 1.96^2 can be considered statistically significant and potentially worthwhile, and a reduction of 9 or more might be considered convincing.
Nonlinear versus linear effects	A linear effect implies a constant relationship between an explanatory variable and the response variable across the range of the explanatory variable, whereas nonlinear effects indicate a more complex relationship. For example, the relationship between age and death in men is increased in early adulthood (from risk-taking behaviour) before reducing in middle years and then increases in old age.
Transformation	Data transformation is the replacement of a variable by a function of that variable. This is commonly used to address non-linear effects. For example, if the variable follows an exponential distribution model fit may be improved by taking the log(e) of time. The appropriateness of a transformation can be examined via improvement of the AIC.
Model reduction techniques	Model reduction techniques are used to reduce the complexity of a generalized model by including only those explanatory variables which improve model fit. Such techniques can include forward, backward, and stepwise. A variant is to include 1 or more explanatory variables regardless of the impact on model fit (e.g. when a relationship is well established in the scientific literature) and only use model reduction techniques with those variables for which the contribution is uncertain. Care is required to ensure that the number of explanatory variables is not excessive [e.g. not less than total *n*/10 for a continuous model or the smaller of (the number of events/non-events)/10]. Having too many explanatory variables risks overfitting [14] where there is a chance that relationships in the data set are modelled rather than reflecting the underlying effects.
Spearman's rank correlation	Spearman's rank correlation is a description of the statistical dependence in the ranking of 2 variables (i.e. how close the predicted rankings were to the actual rankings).
Confidence intervals	The confidence intervals (typically 95%) describe the plausible range of the true value estimated by the model. The point estimate is the best estimate, and the likelihood of the true value reduces moving away from the point estimate towards either the upper or lower CI limit.
Coefficient of determination	The coefficient of determination is the proportion of variation in the dependent variable that is predictable from the explanatory variables (i.e. how completely the model can predict the outcome).
TRIPOD guidelines	The TRIPOD guidelines are a checklist of criteria for the reporting of studies developing, validating, or updating a prediction model, whether for diagnostic or prognostic purposes [15].
Model calibration	Model calibration is the extent to which the model predictions are in line with the observed experience. Some models require regular adjustment because the relationships between variables change over time reducing the calibration.
Model discrimination	Model discrimination is the extent to which the prognostic predictions correctly identify the higher and lower risk individuals.

AIC: Akaike's Information Criterion; CI: confidence interval.

into white or non-white to address assertions that black and minority ethnic celebrities exit *SCD* sooner than white contestants [12, 13]. Professional partners were identified individually when they had 7 or more partners across the series, whilst those with fewer than this were collated as a comparator group. Series was included as a classification variable to allow for differences in the series construction (7–13 rounds).

In the updated analysis, average judges' score for the first 2 episodes (i.e. prior to any contestant leaving) were then incorporated as a marker of dance ability alongside demographic data, to test the hypothesis that dance ability is predictive of how well a contestant performs in the competition.

Model generation

We constructed a multivariable prognostic model to predict the order that celebrity participants would leave *SCD*, utilizing the framework by Harrell *et al.* [14]. This seminal text provides a comprehensive framework and technical explanations for many of the issues we address and remains one of the 'go to' articles

Table 2: Comparison of our *Strictly Come Dancing* prognostic model with the risk models used in the adult heart risk-adjusted 30-day, 1-year and 5-year transplant survival [10]

	Adult heart risk survival rate [10]	Strictly Come Dancing prognostic model
Dataset	All patients ≥18 years registered for the first time for a heart or lung transplant between 1 January 2008 and 31 December 2019. Exclusions included unavailability of certain variables, multi-organ transplant, patients not listed prior to transplant, patients first registered on another transplant list, patients registered outside the UK, and adult patients registered at paediatric centre.	All contestants that have appeared on *SCD* between series 1 (2004) and series 19 (2021). No exclusions.
Variables	Donor cause of death, donor BMI, donor age, respiratory arrest, recipient BMI, recipient creatinine at transplant, VAD at transplant, hospital status at transplant, primary disease, sex mismatch, ischaemia (h), OCS used on heart, interaction between ischaemia time and OCS.	Age, sex, profession, ethnicity, professional partner, series, average judges' scores from first 2 weeks (updated model).
Model	Cox constant proportional hazards survival model to predict the expected survival rates at each cardiothoracic centre. A Poisson mixed regression model was then used to compare the observed and expected number of deaths at each unit.	Generalized linear model where the contribution of the explanatory variables to the response variable was added.
Outcome of prediction	Survival time. This was defined as the time from joining the transplant list to death, regardless of the length of time on the transplant list, whether or not the patient was transplanted and any factors associated with such a transplant, e.g. donor type. Survival time was censored at either the date of removal from the list, or at the last known follow-up date post-transplant when no death date was recorded, or at 17 July 2020 if the patient was on the transplant list at the time of analysis.	Position of exit from the competition. The model produces 'model scores' and the uncertainty of the prediction for each contestant. These were then ranked to give an estimate of the likely position of exit from the competition.

SCD: *Strictly Come Dancing*; BMI: body mass index; OCS: organ care system; VAD: ventricular assist device.

for anyone wishing to find out more about this particular area of statistics.

We considered different generalised models including a Poisson Mixed model and a binomial mixed model but opted for a continuous model as alternative approaches added challenges to interpretation without additional advantages. The generalized linear model describes the various relationships between the explanatory variables described above (e.g. age, sex and occupation) and the response variable (the order in which the contestant left the competition). These contributions were added together and the model was then used to predict the results of the 20th series at its inception. The model gives each candidate an overall 'model score' (an indicator of likely success in the competition) and the uncertainty of this prediction. These were then used to predict the position of competition exit. An overview of our model compared with a clinical example from cardiothoracic surgery (adult heart transplant risk-adjusted 30-day, 1-year and 5-year survival models) can be seen in Table 2.

Model fit

We used a number of techniques to assess and improve model fit. We hypothesized that age might have a complex relationship with outcome. Therefore we fitted a restricted cubic spline with 5 knots for age and assessed model fit with Akaike's Information Criterion (AIC), to examine the extent to which age may have non-linear effect, with a significant improvement in fit required to move to the more complex model (e.g. AIC reduction $>1.96^2$) [14]. The AIC, which is a penalized log-likelihood measure of model fit, was not improved through inclusion of the non-linear restricted cubic spline, so age was untransformed in the final models. We used no model reduction techniques because of our conviction that the candidate explanatory variables we included were important, regardless of their predictive value.

Harrell describes the judicious use of model reduction techniques such as stepwise selection to reduce the number of candidate explanatory variables in a model where these add little or nothing to the model fit [14]. However, in circumstances where prior knowledge indicates the importance of a candidate variable it should be included even when it has an inconclusive effect in a model. To avoid overfitting (the inclusion of too many variables and subsequent poor external validity), we limited the number of degrees of freedom for the explanatory variables to <1/10 of the number of observations [14].

Comparison with the actual series results

We compared the 2 predicted rankings of the 2022 contestants (series 20) with the final results using Spearman's rank correlation with exact 95% confidence intervals and also supplied the R squared value or coefficient of determination (R^2). All analyses were conducted in SAS version 9.4 (SAS Institute, Cary, NC). This study was performed in accordance with the TRIPOD guidelines [15].

RESULTS

Across the 19 completed series of *SCD*, there were 267 couples, with 15 pairings for series 20 for whom the model predicted scores. Characteristics are described in Table 3. The median age was 38.3 years, with half of the celebrities being female. The most common profession was actor (76, 27.2%) and 64 (22.5%) celebrities were non-white.

In the baseline model, age was the only variable independently statistically significant, with each additional year reducing the predicted mean score (the round when the contestant might expect to leave *SCD*) by 0.14 [95% confidence interval (CI) 0.10–0.18, $P < 0.0001$]. Ethnicity was near significant, however, with non-white participants receiving a predicted score 0.99 lower than white participants (95% CI −2.07 to 0.08, $P = 0.07$). In terms of career, politicians performed worse than singers/musicians and actors, but better than other broadcasters, sportspeople, comedians, other professionals and model/fashion professions. Although not statistically significant, partnership with Giovanni Pernice was associated with a 2.13 increase in expected score (95% CI −0.68 to 4.93, $P = 0.14$), and Janette Manrara a reduction of 1.03 (95% CI −3.70 to 1.63, $P = 0.44$).

Table 3: Baseline characteristics of contestants used as prognostic factors

Characteristic	Value ($n = 279$)
Female, *n* (%)	140 (50.2%)
Age, median (IQR)	38.3 (31.5, 48.4)
Physical disability, *n* (%)	7 (2.5%)
Non-white, *n* (%)	64 (22.9%)
Profession, *n* (%)	
Actor	76 (27.2%)
Other broadcaster	73 (26.2%)
Sportsperson	49 (17.6%)
Singer/musician	38 (13.6%)
Other professional	19 (6.8%)
Comedian	13 (4.7%)
Model/fashion industry	7 (2.5%)
Politician	4 (1.4%)
Professional partner, *n* (%)	
Anton du Beke	18 (6.5%)
Brendan Cole	15 (5.4%)
Erin Boag	10 (3.6%)
Ola Jordan	10 (3.6%)
Karen Hauer	10 (3.6%)
Aljaž Skorjanec	9 (3.2%)
Janette Manrara	8 (2.9%)
James Jordan	8 (2.9%)
Pasha Kovalev	8 (2.9%)
Kristina Rihanof	8 (2.9%)
Giovanni Pernice	8 (2.9%)
Katya Jones	7 (2.5%)
Flavia Cacace	7 (2.5%)
Natalie Lowe	7 (2.7%)
Vincent Simone	7 (2.7%)
Other	139 (49.8%)

IQR: interquartile range.

Model predicted scores are described in Fig. 1 and Table 4. In the baseline model, R^2 was 0.25. Following initial analysis, the multivariable prognostic model was rerun to include the mean score awarded by the judges for the first 2 weeks to represent the contestants actual dance ability. This increased the R^2 to 0.40. Molly Rainford was predicted to win the series in both iterations, whilst Kaye Adams was predicted to leave first in the initial model and Tony Adams in the update. Kaye Adam did end up leaving first, but the series was won by Hamza Yassin. The Spearman rank correlation for the baseline model (Rho) was 0.22 (95% CI −0.33 to 0.66; $P = 0.42$) and for the model updated with the judges' scores 0.30 (95% CI −0.25 to 0.70; $P = 0.27$).

DISCUSSION

SCD is a popular BBC television programme that started in the UK and has spread internationally. The potential outcome of each series is a topic of debate in the media, although this study's authors and bookmakers alike have struggled to predict the results: 2020 champions Bill Bailey and Oti Mabuse started as 50/1 outsiders, whilst Aston Merrygold and Janette Manrana started as 2017 favourites at 11/4 but suffered surprise elimination in week 7 [16].

We created a prognostic model attempting to predict contestants' performance-based purely on demographic data, and an updated model including the average judging score from the first 2 episodes. Despite limited prognostic benefit (R^2 = 0.25), the baseline model was not without merit. In 2022, the lowest

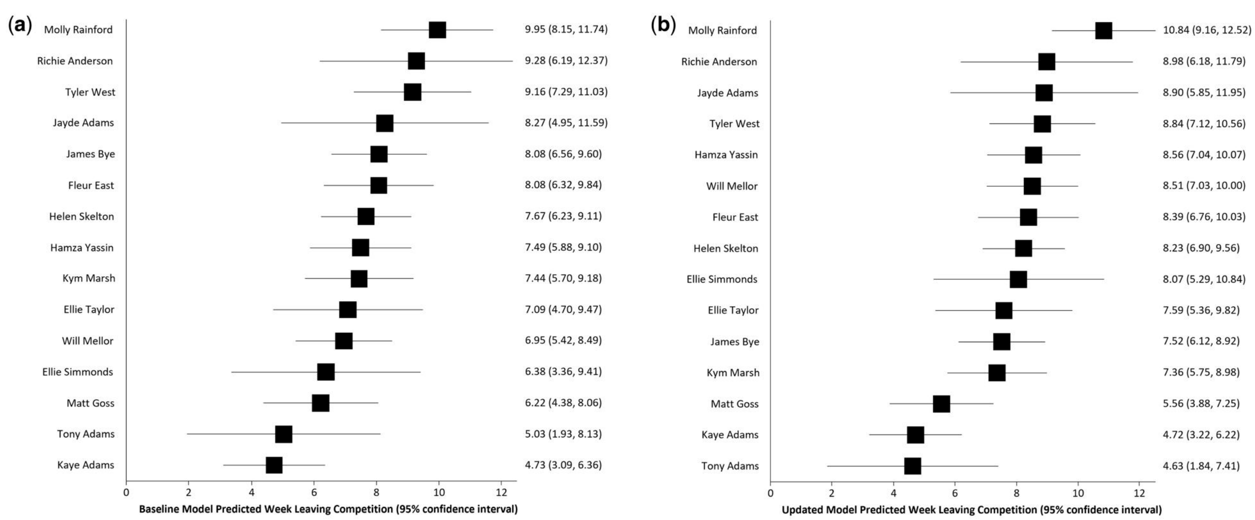


Figure 1: Predicted scores and 95% confidence interval for participants in series 20 (2022 series). (**a**) Baseline score, R^2 0.25. (**b**) Updated score incorporating mean scores for the first 2 weeks, R^2 0.40.

Table 4: Predicted outcomes for participants in series 20 (2022 series)

Contestant	Partner	Actual order	Baseline model score	95% CI		Predicted order (baseline score)	2-Week scores	Updated model score	95% CI		Predicted order (updated score)
				Baseline lower	Baseline upper				Updated lower	Updated upper	
Hamza Yassin	Jowita Przystal	1	7.49	5.88	9.10	8	58	8.56	7.04	10.07	5
Fleur East	Vito Coppola	=2	8.08	6.32	9.84	6	57	8.39	6.76	10.03	7
Helen Skelton	Gorka Marquez	=2	7.67	6.23	9.11	7	53	8.23	6.90	9.56	8
Molly Rainford	Carlos Gu	=2	9.95	8.15	11.74	1	65	10.84	9.16	12.52	1
Will Mellor	Nancy Xu	5	6.95	5.42	8.49	11	60	8.51	7.03	10.00	6
Kym Marsh	Graziano Di Prim	6	7.44	5.70	9.18	9	50	7.36	5.75	8.98	12
Ellie Taylor	Johannes Radebe	7	7.09	4.70	9.47	10	51	7.59	5.36	9.82	10
Tyler West	Dianne Buswell	8	9.16	7.29	11.03	3	53	8.84	7.12	10.56	4
Tony Adams	Katya Jones	9	5.03	1.93	8.13	14	37	4.63	1.84	7.41	15
Ellie Simmonds	Nikita Kuzmin	10	6.38	3.36	9.41	12	56	8.07	5.29	10.84	9
James Bye	Amy Dowden	11	8.08	6.56	9.60	5	46	7.52	6.12	8.92	11
Jayde Adams	Karen Hauer	12	8.27	4.95	11.59	4	49	8.90	5.85	11.95	3
Matt Goss	Nadiya Bychkova	13	6.22	4.38	8.06	13	42	5.56	3.88	7.25	13
Richie Anderson	Giovanni Pernice	14	9.28	6.19	12.37	2	55	8.98	6.18	11.79	2
Kaye Adams	Kai Widdrington	15	4.73	3.09	6.36	15	43	4.72	3.22	6.22	14
Spearman rank correlation baseline model				−0.33	0.66	0.22 ($P = 0.42$)	Spearman rank correlation updated model		−0.25	0.70	0.30 ($P = 0.27$)

The final score relates to judging week when the contestant left the competition. The baseline score relates to the contestant characteristics model whilst the updated score relates to the baseline model with incorporated mean 2-week scores.

CI: confidence interval.

scored subject left *SCD* first. However, and critical to this study, it is notable that explanatory variables included (i.e. age, sex, profession, ethnicity, physical disability and professional partner) do not directly relate to whether the contestant can dance. The updated model, which included the average judges' scores for the first 2 episodes, improved the model's performance and prediction of the true results, although both provided only weak correlation and neither model was established as better than chance.

A strength of our model is that the results can be validated and updated each year, with new series providing new rankings of contestants. All models will lose validity over time as populations shift and clinical circumstances change and hence have to be continually updated. External validation is important in prognostic models and the arrival of new data each year avoids difficult decisions on how much data to hold back from model building to have an adequate validation. In fact, our data set is rather sparse, although sample size considerations are rather limited in practice for prognostic model building with the exception of the avoidance of over fitting. A useful statistical technique to evaluate the internal validity of models which necessarily utilize all available data in the initial build has been proposed by Harrell *et al.* [14] and we recommend readers to consider such internal validation approaches particularly where data are sparse.

A further strength of our approach, where we are attempting to predict the ranking of contestants, is that we are not subject to questions regarding calibration (the degree to which an additional risk predicted is realized in practice). By contrast, models including EuroSCORE have required regular calibration to improve model prediction [17].

Limitations

A fundamental limitation of this study is the lack of a component describing contestant popularity. In the updated version, Tony Adams was predicted to leave first and scored lowest after 2 weeks. Despite this, he neither left first nor was in the dance-off demonstrating more public support than expected. Similarly, the model substantially overestimated Richie Anderson and Jayde Adams' performances, who left much earlier than predicted. Indeed, the lack of information on the public's voting behaviour in the model substantially limits its predictive ability. One option would be to include a parameter from participants' social media following, however as this is unlikely to be comparable across the 20 series, this option was not pursued.

A further limitation of our updated model is that, although 2 weeks of judges' scores captures a starting point, celebrities who improve substantially over the series may have their underlying dance ability poorly captured. Updating the model to reflect likelihood of leaving based on later judges' scores or on whether contestants were in the dance-off (associated with popularity) would become tautological, as a celebrity must be in the dance-off to leave, and celebrities who remain on average increase their scores over time. It is interesting to note the increased clustering of predicted values in the second model, indicating that the field is relatively open, and much can change in later weeks.

It should also be noted that due to the limited number of *SCD* series, only 279 participant datasets were used for the model. This compares to 19 030 for the original description of EuroSCORE and 774 881 for STS [4–6].

It should also be considered whether it is necessary and appropriate to develop a new risk prediction model to answer the clinical or research question under consideration. As discussed above, the data used to generate the model must be of appropriate magnitude and quality, but also the parameters selected should be able to be easily acquired from the population it is applied to in a typical clinical context. An alternative is to update a pre-existing model as described above.

Finally, our statistical model is one of a number of approaches to predicting outcome. Some of these have been tried, but the paper's message remains qualitatively the same. There may also be a role for neural networks for survival prediction in clinical medicine [18]. Although not widespread at present, such approaches will doubtless become more common as research in the area develops.

Wider impact on cardiothoracic surgery and beyond

Whilst not all readers may share our enthusiasm for *SCD*, the danger of using prognostic models developed from insufficiently useful demographic factors is not limited to the world of television. All models must include appropriately explanatory variables for them to act as effective risk assessment tools. For example, in the assessment of heart and lung transplant services in the UK, several factors are included in the prognostic risk model for survival rates, although the only patient severity items are body mass index (BMI), creatinine as a marker of renal function, ventricular assist device at transplant, in or out of hospital, and their primary cardiac disease (Table 2) [10]. Beyond these broad categories, there is no variable to identify how sick a patient is. Similar to this model's inability to address a contestant's dance ability and popularity, for prognostic models of heart and lung transplant it is necessary to include descriptive factors of donor and recipient health to compare services appropriately. Models that do not include this information run the risk of incorrectly identifying a centre as performing badly when they simply intervene on more unwell patients. Such errors of construction risk the gaming of scores, e.g. avoiding intervention on higher-risk individuals for whom the procedure could be potentially very beneficial because of the increased risk of a poor outcome even though the risk–benefit ratio for the individual may indicate that the procedure is very worthwhile. Such perverse incentives may be considered a form of moral hazard.

Criticisms have been levelled at other models in cardiothoracic surgery, which perform variably across different patient populations. In a 2016 study comparing EuroSCORE, EuroSCORE II and STS in an American population, all models were found to overestimate the risk of death, EuroSCORE most markedly, with its successor and STS having similar predictive value (predicted mortality 7.8%, 3.3% and 2.7% respectively with 1.8% observed deaths) [18]. However, this prediction varies in different patient subcategories. Pooled data from the Netherlands and the UK found a poor predictive value in patients deemed high risk in both EuroSCORE and EuroSCORE II [19]. Furthermore, in this research, there was no improvement between earlier and later versions. One of the suggestions for difference across age was a lack of description of patient frailty which has previously been identified as a key risk factor for cardiac surgery [20]. A more recent study in a Swedish population identified a general overestimation of risk across all risk stratifications, although found this was more pronounced in those deemed lower risk [21]. They also found worse calibration in younger age groups.

The difference in risk prediction across these studies is not necessarily surprising and could be accounted for by changes in healthcare delivery between studies or differences in patient populations across geography. The original EuroSCORE model was based on data from 128 surgical centres across 8 European countries, and model discrimination varied between different regions (from 0.74 in Spain to 0.87 in Finland) [4, 22]. EuroSCORE II consisted of an even more geographically diverse population, covering a total of 154 centres from 43 countries (including 27 European countries and 16 non-European countries, including Argentina, Sudan and Taiwan). Healthcare delivery is likely very different in these countries, and the results of grouping these patients may mean they are less relatable to a patient from a specific country. Similarly, in our model, the contestants used to generate our data set do not necessarily come from the same population as those we were predicting for, and the results cannot necessarily be applied between the groups.

It should be noted that our model has not yet been independently externally validated. External validation is crucial to determining a model's reproducibility and generalizability, and it is therefore important that this is carried out prior to the implementation of a model in practice [23]. In a recent systematic review of models developed for COVID-19 diagnosis and prognosis, the authors demonstrated that only 5% of the 232 included were externally validated by a calibration plot, significantly limiting confidence in their use [24]. The authors also demonstrated a high risk of bias (including enrolling participants who were unrepresentative of the models' targeted populations, model overfitting and unclear reporting) across studies. These issues are not limited to COVID-19 research and have been identified in the use of prognostic models across medicine [25, 26]. To address concerns that the reporting of prediction models was generally poor, the TRIPOD group produced a checklist of criteria to be included deemed essential [15]. This should be adhered to and accompany all submitted studies, as was the case with the present report.

CONCLUSION

The results of *SCD* can be predicted using baseline characteristics to a limited degree, but this prediction improves substantially when including empirical data on whether contestants can actually dance. In relation to cardiothoracic surgery and other medical sciences, clinicians and researchers should be cautious when drawing conclusions or developing predictive models from limited data where these alone do not capture the prediction of interest (e.g. how sick a particular patient might be).

Finally, all prognostic models merely provide risk estimates. It is therefore still perfectly possible that a patient with high perceived risk has an excellent surgical outcome whilst the opposite may be true for a low-risk patient. However, predictive they are on paper, the output of these tools must be interpreted in the context of all patient and surgical factors before they can be applied to the person sat in front of their surgeon considering a life-changing (for better or for worse) operation.

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Data are derived from freely available online information (including but limited to Wikipaedia https://www.wikipedia.org/). Our compiled dataset is available on request.

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Author contributions

Jamie A. Mawhinney: Conceptualization; Data curation; Investigation; Methodology; Project administration; Resources; Writing—original draft; Writing—review & editing. **Craig A. Mounsey:** Data curation; Investigation; Writing—review & editing. **Alastair O'Brien:** Investigation; Methodology; Writing—review & editing. **J. Rafael Sádaba:** Supervision; Visualization; Writing—review & editing. **Nick Freemantle:** Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Supervision; Validation; Visualization; Writing—review & editing.

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